C Programming Questions and Answers

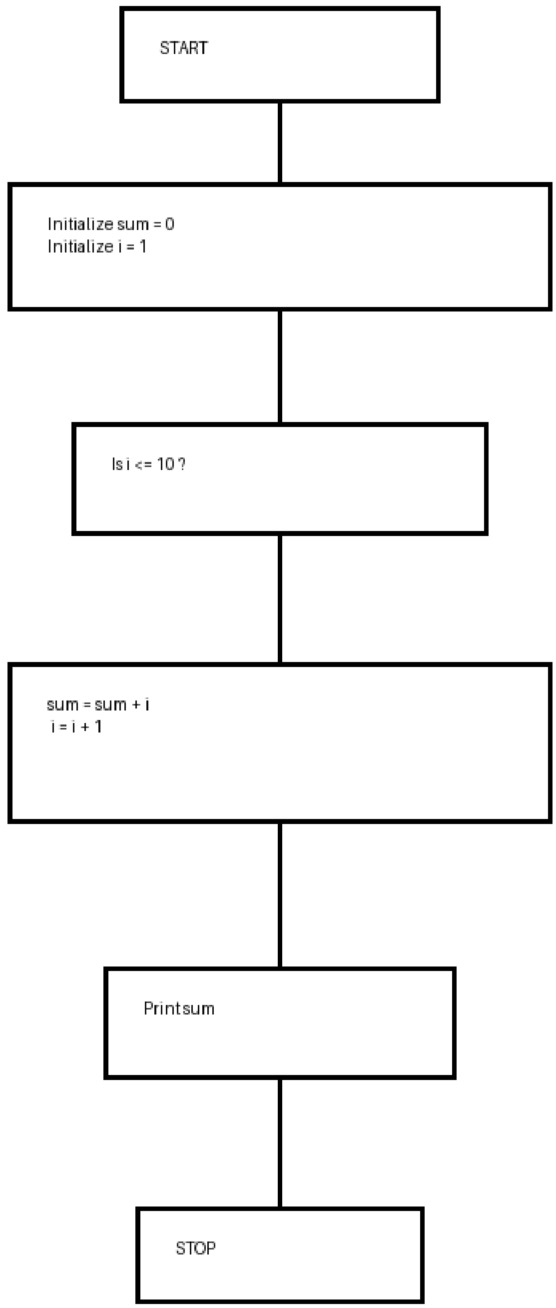
1. Difference between while and do-while loop

while loop	do-while loop
Condition is checked before execution of loop body.	Condition is checked after execution of loop body.
May execute zero times if condition is false.	Executes at least one time.
Entry controlled loop.	Exit controlled loop.
No semicolon after condition.	Semicolon is used after condition.

2. Algorithm and Flowchart for Summation of First 10 Positive Numbers

```
Detailed Algorithm:

Step 1: Start the program.

Step 2: Declare three integer variables:
        i  -> used as loop counter
        sum -> used to store the total sum

Step 3: Initialize:
        sum = 0
        i = 1

Step 4: Check the condition whether i <= 10.

Step 5: If the condition is true:
        Add the value of i to sum.
        sum = sum + i

Step 6: Increase the value of i by 1.
        i = i + 1

Step 7: Repeat Steps 4 to 6 until i becomes greater than 10.

Step 8: Display the final value of sum.

Step 9: Stop the program.

Output:
The summation of first 10 positive numbers is 55.
```

Flowchart:


START
Initialize sum = 0
Initialize i = 1
Is i <= 10 ?
sum = sum + i
i = i + 1
Print sum
STOP

3. Different Types of Operators in C Language

```
1. Arithmetic Operators
Used for mathematical calculations.
Examples: +, -, *, /, %

2. Relational Operators
Used to compare values.
Examples: ==, !=, >, <, >=, <=

3. Logical Operators
Used to combine conditions.
Examples: &&, ||, !

4. Assignment Operators
Used to assign values.
Examples: =, +=, -=, *=, /=

5. Increment and Decrement Operators
Examples: ++, --

6. Bitwise Operators
Examples: &, |, ^, ~, <<, >>

7. Conditional Operator
Syntax:
condition ? expression1 : expression2;
```

4. C Program to Print First 10 Even Numbers Using for Loop

```c
#include <stdio.h>

int main()
{
    int i;

    printf("First 10 Even Numbers:\n");

    for(i = 2; i <= 20; i = i + 2)
    {
        printf("%d\n", i);
    }

    return 0;
}
```

5. C Program to Print First 10 Odd Numbers Using while Loop

```c
#include <stdio.h>

int main()
{
    int i = 1, count = 0;

    while(count < 10)
    {
        printf("%d\n", i);
        i = i + 2;
        count++;
    }

    return 0;
}
```

6. C Program to Check Whether a Number is Prime or Not

```
#include <stdio.h>

int main()
{
    int n, i, flag = 0;

    printf("Enter a number: ");
    scanf("%d", &n);

    if(n <= 1)
    {
        printf("Not a Prime Number");
    }
    else
    {
        for(i = 2; i <= n/2; i++)
        {
            if(n % i == 0)
            {
                flag = 1;
                break;
            }
        }

        if(flag == 0)
            printf("Prime Number");
        else
            printf("Not a Prime Number");
    }

    return 0;
}
```

7. C Program to Check Whether an Alphabet is Vowel or Not Using switch-case

```
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter an alphabet: ");
    scanf("%c", &ch);

    switch(ch)
    {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
            printf("It is a Vowel");
            break;

        default:
            printf("It is not a Vowel");
    }

    return 0;
}
```